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Track Name: Artificial Intelligence and Data Science

Paper ID: 871

Paper Title: Student Academic Performance Prediction Using Logistic Regression and Random Forest with SHAP-Based Interpretability

Abstract:
Academic failure seldom comes out of the blue—there are indications in attendance records, late assignments, and decreasing participation well before a failing mark. This research paper investigates the possibility of using machine learning classification of institutional data collected regularly to identify with reasonable reliability and interpretability academically at-risk students early. The UCI Student Performance dataset (395 records, Portuguese secondary school mathematics) was used to evaluate two supervised classifiers, namely Logistic Regression, as an interpretable baseline and Random Forest, as an ensemble method. They both had high test performance with 92.4% but Random Forest had better Fail-class recall (0.96 vs. 0.93), AUC (0.95) and cross-validation stability (mean 90.6%, 0.022 SD). Interpretability based on SHAP verified that G2 and G1 are the strongest predictors and the secondary indicators were prior failures and absenteeism, both of which are easily accessible to institutions on a regular basis. An empirical three-stage prediction-prioritization-response model is suggested to be practically implemented.

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